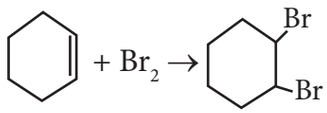
Evaluation

I. Choose the best answer.

1. For the following reactions

(A) $CH_3CH_2CH_2Br + KOH \rightarrow CH_3–CH= CH_2 + KBr + H_2O$

(B) $(CH_3)_3CBr + KOH \rightarrow (CH_3)_3 COH + KBr$

(C)  $+ Br_2 \rightarrow$

Which of the following statement is correct?

(a) (A) is elimination, (B) and (C) are substitution

(b) (A) is substitution, (B) and (C) are elimination

(c) (A) and (B) are elimination and (C) is addition reaction

(d) (A) is elimination, B is substitution and (C) is addition reaction.

2. What is the hybridisation state of benzyl carbonium ion?

(a) sp^2 (b) spd^2
(c) sp^3 (d) sp^2d

3. Decreasing order of nucleophilicity is

(a) $OH^- > NH_2^- > {}^-OCH_3 > RNH_2$
(b) $NH_2^- > OH^- > {}^-OCH_3 > RNH_2$
(c) $NH_2^- > CH_3O^- > OH^- > RNH_2$
(d) $CH_3O^- > NH_2^- > OH^- > RNH_2$

4. Which of the following species is not electrophilic in nature?

(a) Cl^+ (b) BH_3 (c) H_3O^+ (d) $^+NO_2$

5. Homolytic fission of covalent bond leads to the formation of

(a) electrophile (b) nucleophile
(c) Carbo cation (d) free radical

6. Hyper Conjugation is also known as

(a) no bond resonance
(b) Baker - nathan effect
(c) both (a)and (b)
(d) none of these

7. Which of the group has highest +I effect?

(a) CH_3- (b) CH_3-CH_2-
(c) $(CH_3)_2$-CH- (d) $(CH_3)_3$-C-

8. Which of the following species does not exert a resonance effect?

(a) C_6H_5OH (b) C_6H_5Cl
(c) $C_6H_5NH_2$ (d) $C_6H_5\overset{+}{N}H_3$

9. -I effect is shown by

(a) -Cl (b) -Br
(c) both (a) and (b) (d) $-CH_3$

10. Which of the following carbocation will be most stable?

(a) Ph_3C^- (b) $CH_3-\overset{+}{C}H_2-$
(c) $(CH_3)_2-\overset{+}{C}H$ (d) $CH_2= CH - \overset{+}{C}H_2$

11. Assertion: Tertiary Carbocations are generally formed more easily than primary Carbocations ions.

Reason: Hyper conjugation as well as inductive effect due to additional alkyl group stabilize tertiary carbonium ions.

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